HeadGenome: Dynamic Entropy-Collapse Probing Does Not Isolate Causal Attention Circuits
A Multi-Architecture Causal and Geometric Audit

Anonymous ACL submission

Abstract

Unsupervised diagnostics that classify attention heads into functional taxa—such as retrieval heads, induction heads, and attention sinks—promise scalable circuit discovery without costly task-specific supervision. In this work, we present a causal and geometric audit of dynamic task-contrast entropy probing across 1,568 attention heads in three model families (GPT-2-medium, Qwen-2.5, and LLaMA-3.2), with systematic causal interventions across 720 heads in GPT-2 and Qwen-2.5. We demonstrate that dynamic entropy probing fails to isolate causal circuits: (1) Heads identified as retrieval by entropy collapse are causally inert on long-context Needle-in-a-Haystack (NIAH: 100% accuracy preserved under ablation), whereas true needle-routing heads collapse accuracy to 37.0% in Qwen-2.5-0.5B; evaluated against continuous needle-routing attention, the probe achieves an AUC-ROC of only 0.43 [0.21, 0.65], consistent with chance, identifying zero causal knockout heads (0.0% sensitivity [0.0%, 49.0%]). In GPT-2, retrieval exhibits high circuit redundancy (95.0% vs 100.0%), showing that no retrieval subset is causally necessary there. (2) Standard synthetic NIAH conflates retrieval with prefix-matching induction due to verbatim prompt repetition, explaining why Olsson heads collapse NIAH (3.0% in Qwen, 0.0% in GPT-2). (3) For induction, the probe partially correlates with prefix scores (AUC 0.790) but heavily dilutes the circuit (10.2% Jaccard overlap with Olsson heads), recovering only +17.8% on IOI (vs +67.7% for Olsson heads) and +0.29 nats prefix loss (vs +1.84 nats). (4) Perplexity shifts fail to detect factual retrieval collapse (+0.24 PPL despite a 63% accuracy drop in Qwen). (5) Weight Frobenius ratios reflect architectural depth gradients rather than functional roles. We conclude that dynamic entropy probing captures statistical recurrence geometry rather than causal mechanisms, and release our unconfounded audit suite.

1 Introduction

A central aspiration of mechanistic interpretability is reverse-engineering the algorithmic sub-circuits implemented by multi-head self-attention (Elhage et al., 2021). Prior work has identified specialized functional units: *induction heads* that drive in-context few-shot replication (Olsson et al., 2022), *attention sinks* that absorb surplus softmax mass (Xiao et al., 2024), and *retrieval heads* that copy distant tokens into the residual stream (Wu et al., 2024).

Task-contrast attention sharpening is an intuitive diagnostic for unsupervised circuit discovery: if an attention head's softmax entropy sharply decreases upon presenting a factual query relative to a neutral prompt ($\Delta H > 0$), it is natural to hypothesize that the head specializes in factual retrieval. Similar entropy-based metrics have been explored as indicators of head confidence and specialization (Clark et al., 2019), while factual recall circuits have been isolated via direct causal interventions (Geva et al., 2023). In this work, we formalize this intuitive premise as the **dynamic task-contrast probe** (Δ-probe) and conduct a rigorous causal stress test across 1,568 attention heads in GPT-2-medium (Radford et al., 2019), Qwen-2.5 (Yang et al., 2024), and LLaMA-3.2 (Dubey et al., 2024), with systematic causal interventions across 720 heads in GPT-2 and Qwen-2.5. We formulate the core mechanistic question: *Can prompt-contrast attention sharpening serve as an unsupervised proxy for causal circuit discovery, or does it conflate statistical entropy dynamics and architectural depth gradients with causal circuit execution?*

Our contributions are as follows:

- **Causal Dissociation in Retrieval:** Probed retrieval heads (layers 1, 4, 6 in Qwen; lay-

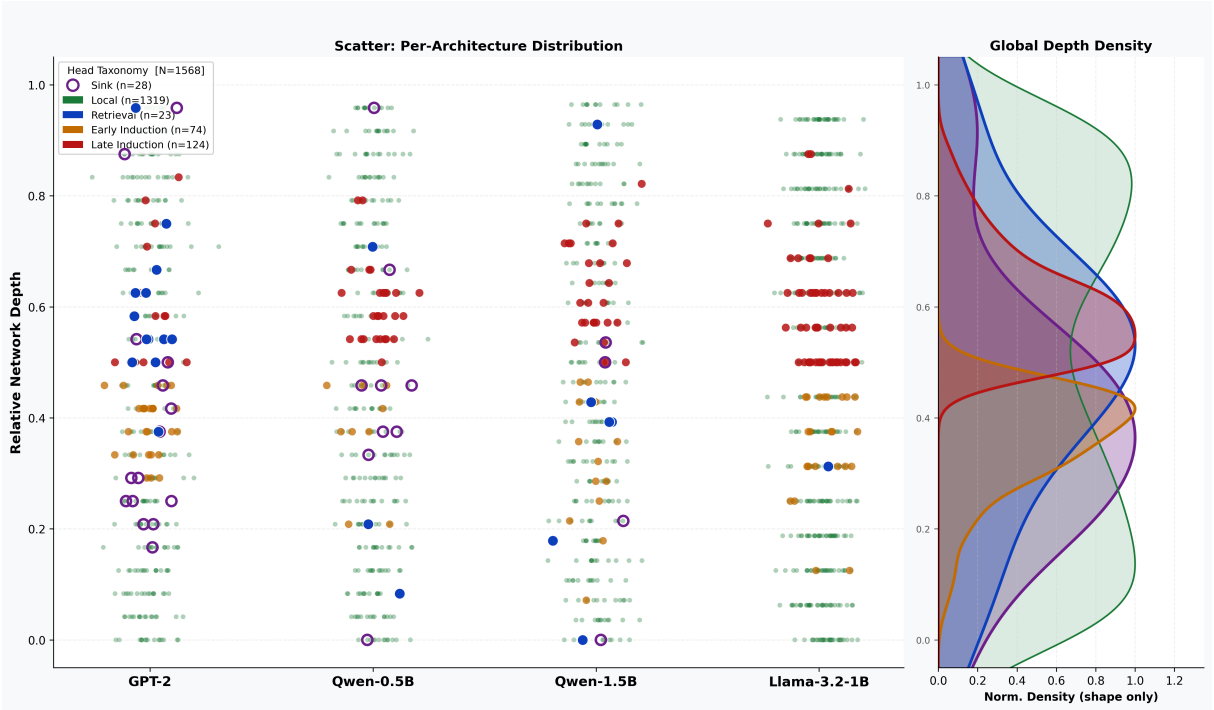


Figure 1: Taxonomic Attention Head Map: Spatial distribution of functional attention head types across transformer architectures ($N = 1,568$ heads across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B). This taxonomy serves as the empirical substrate for the causal and geometric audit in this paper. Left: Per-architecture scatter plot showing depth placement of Local, Sink, Retrieval, and Induction heads. Right: Global depth density curves normalized per class.

ers 1, 3, 5, 11, 14, 15, 18, 23 in GPT-2) are causally inert on Needle-in-a-Haystack (NIAH: $100.0\% \rightarrow 100.0\%$). True needle-routing heads in Qwen reside in late layers (15–19), where ablating the top 4 heads (calibrated on 20 trials, evaluated on 100 disjoint trials) collapses accuracy to 37.0% $[28.2\%, 46.8\%]$ (while 20 draws of layer-matched late controls retain $97.6\% \pm 1.4\%$). In GPT-2, retrieval routing exhibits high circuit redundancy (95.0% preserved under 13-head ablation), showing that no localized retrieval set is causally necessary there. Evaluated against continuous needle-routing attention, the probe achieves an AUC of 0.43 $[0.21, 0.65]$ (consistent with chance) and 0.0% sensitivity $[0.0\%, 49.0\%]$ on causal heads.

- **Induction Circuit Dilution:** Decoupling circuit identity from layer depth via activation patching on IOI and standard held-out repeated-token loss demonstrates that canonical Olsson heads recover $+67.7\%$ logit difference and drive a $+1.84$ nat loss degradation on repeated prefixes, compared to $+17.8\%$ and $+0.29$ nats for probed induction heads (and $+1.2\%$ / $+0.09$ nats for layer-matched late controls). While the probe correlates with prefix scores (AUC 0.790 in Qwen, 0.690 in GPT-2), it shares only 10.2% Jaccard overlap with Olsson heads, heavily diluting the causal circuit with inert units.
- **The Perplexity Illusion:** Ablating causal retrieval heads collapses NIAH accuracy by 63.0% while shifting WikiText-2 PPL by only $+0.24$ ($20.01 \rightarrow 20.25$, an excess of $+0.21$ over late random controls), whereas probed retrieval heads cause $+0.27$ with zero accuracy drop. Relative to capacity controls, general induction ablation incurs an excess PPL penalty of $+1.73$ in Qwen and $+4.13$ in GPT-2.
- **Weight Frobenius Norm Ratios:** Frobenius norm ratios $\|W_V\|_F / \|W_Q\|_F$ are dominated by network depth ($r = 0.635$–0.734), while within-layer correlations with functional identity are either null (GPT-2, Qwen-0.5B: $r \approx 0.00$) or weak ($r = 0.11$–0.19 in Qwen-1.5B and LLaMA-3.2, explaining $< 3.5\%$ of variance).

- **Methodological Audit & Diagnostic Insights:** We reveal that standard synthetic NIAH tasks act as prefix-matching induction tasks due to verbatim prompt repetition. We resolve the random control choke hypothesis, establish a mass-based sink criterion ($> 90\%$ BOS mass / $H < 0.50$ bits, identifying 70 sinks in LLaMA), and evaluate GQA query head independence against a layer-conditional permutation null (50.8% mixed groups observed vs 68.9% expected with sinks, $p < 10^{-4}$). We open-source our audit suite at `https://anonymous.4open.science/r/headgenome-mechinterp`.

2 Related Work

Mechanistic Interpretability & Circuit Discovery. Mechanistic interpretability aims to reverse-engineer transformers into human-understandable computational graphs (Elhage et al., 2021). Foundational work demonstrated that two-layer attention-only models develop *induction heads* that perform $[A][B] \ldots [A] \rightarrow [B]$ in-context copying (Olsson et al., 2022). In larger autoregressive models, activation patching has been used to isolate specific circuits, such as the Indirect Object Identification (IOI) circuit in GPT-2 small (Wang et al., 2023) and factual recall circuits in medium-scale language models (Geva et al., 2023). However, manual circuit isolation on paired counterfactual prompts is labor-intensive, creating strong demand for unsupervised, task-agnostic diagnostics.

Long-Context Retrieval & Attention Sinks. As context windows expanded to tens of thousands of tokens, attention heads specialized in distant token retrieval were discovered (Wu et al., 2024; Kamradt, 2023). Concurrently, Xiao et al. (2024) observed that initial tokens (e.g., <bos>) consistently receive disproportionate attention mass even when semantically uninformative. These *attention sinks* serve as numeric dump basins for softmax normalization, preventing representations from destabilizing when streaming past cache boundaries.

Head Pruning & Bottleneck Choking. Early structural analyses of multi-head attention established that a large fraction of attention heads can be pruned at test time without substantial loss in BLEU or perplexity (Voita et al., 2019; Michel et al., 2019). However, modern architectures with fewer, wider heads per layer or grouped-query attention (Dubey et al., 2024; Yang et al., 2024) exhibit different failure modes: while pruning dispersed heads across layers is tolerated, concentrating knockouts in specific layers induces representation rank bottlenecks and capacity starvation (Bhojanapalli et al., 2020). Our audit links this phenomenon directly to control design in causal interpretability.

3 Diagnostic Probing Setup

3.1 Method vs. Labels in Entropy Probing

Dynamic entropy probing evaluates attention concentration shifts. In our diagnostic framework, the prompt-contrast probe evaluates attention entropy on a target match prompt versus a neutral distractor prompt at a fixed context length L:

$$H(A_{t,:}^{(l,h)}) = -\sum_{j=1}^{t} A_{t,j}^{(l,h)} \log_2 \left(A_{t,j}^{(l,h)} + \epsilon \right) \quad (1)$$

$$\Delta H^{(l,h)} = H(A_{\text{neutral}}^{(l,h)}) - H(A_{\text{match}}^{(l,h)}) \quad (2)$$

where $A_{t,:}^{(l,h)}$ denotes the attention distribution of head h in layer l at generation position t. Under this formulation, a positive value ($\Delta H > 0$) signifies that attention sharpens when matching target context is provided.

The Cross-Length Entropy Confound. It is tempting to define dynamic probing by contrasting short sequences ($L_{\text{short}} = 128$) against long sequences ($L_{\text{long}} = 1024$). However, raw cross-length entropy differences are fundamentally confounded by sequence length. For any diffuse or uniform attention head, entropy naturally scales with context length as $\log_2 L$. Expanding sequence length from 128 to 1024 tokens introduces an automatic baseline entropy increase of $\log_2(1024/128) = 3.0$ bits. Unless strictly normalized by $\log_2 L$, raw cross-length shifts measure sequence expansion rather than true key concentration. Therefore, unconfounded probing must compare matched-length prompts.

Canonical Taxonomy Thresholds. Using prompt-contrast entropy ΔH at fixed length, heads are assigned to mutually exclusive categories:

- **Attention Sink:** Evaluated first; heads directing near-exclusive attention mass to token 0: $H(A_{\text{match,BOS}}^{(l,h)}) < 0.10$ bits.

- **Retrieval Candidate:** Attention sharpens on match context: $\Delta H^{(l,h)} \geq +0.30$ bits.
- **Induction Candidate:** Attention broadens across candidate tokens: $\Delta H^{(l,h)} \leq -0.50$ bits.
- **Local Candidate:** Remaining heads: $-0.50 < \Delta H^{(l,h)} < +0.30$ bits.

Because retrieval requires $\Delta H \geq +0.30$ and induction requires $\Delta H \leq -0.50$, these classes are mutually exclusive by definition.

3.2 LLaMA Sink Threshold Artifact

In LLaMA-3.2-1B, applying the rigid < 0.10 bits threshold yielded exactly 0 sink heads (Figure 2). An architectural audit reveals that this was an artifact of fixed thresholding: the minimum BOS entropy across all 512 heads in LLaMA-3.2-1B is 0.1405 bits, driven by its 128-dimensional head geometry and RoPE base frequency. Rather than applying a fixed threshold across architectures or relying on an arbitrary percentile cut (which automatically labels 26 heads in any 512-head model), we establish a mass-based criterion: heads directing $\geq 90\%$ of attention mass to token 0 ($A_{\text{BOS}} \geq 0.90$), corresponding to $H(A_{\text{match,BOS}}) < 0.50$ bits. In LLaMA-3.2-1B, this mass-based criterion identifies 70 sink heads (13.7% of all 512 query heads), concentrated primarily in early layers (layers 0–5), correctly resolving the zero-sink artifact while avoiding vacuous percentile definitions.

3.3 GQA Query Head Independence

Grouped-Query Attention (GQA) shares key and value projection matrices across groups of query heads. We tested whether GQA forces sibling query heads to share functional specializations. Across all 128 GQA groups in LLaMA-3.2-1B (32 query heads, 8 KV heads; 4 query heads per group), under the mass-based sink definition ($H < 0.50$ bits, 70 sinks), **65 of 128 groups (50.8%)** contain query heads assigned conflicting behavioral labels. To evaluate whether this reflects true query head specialization or merely layer-level base rates, we evaluate a layer-conditional permutation test ($B = 100,000$ draws): shuffling head labels strictly within each layer yields an expected mixed group fraction of $68.9\% \pm 2.4\%$ (88.3 ± 3.1 groups, $p < 10^{-4}$, $Z = -7.43$). Restricting the taxonomy to non-sink classes likewise yields **35 of 128 mixed groups (27.3%)** versus an expected null of $38.0\% \pm 1.8\%$ (48.7 ± 2.3 groups, $p < 10^{-4}$). Under both formulations, while sibling query heads show statistically significant clustering towards similar behavioral roles ($p < 10^{-4}$), over a quarter to half of all groups assign conflicting specializations to sibling heads sharing identical KV projections. Furthermore, within-group ΔH variance accounts for **33.6%** of total ΔH variance across the entire model. Because each query head has an independent linear projection W_Q, query heads attend to distinct patterns despite shared keys and values.

4 Causal Verification

4.1 Knockout Necessity on Long-Context Retrieval

We execute causal interventions via PyTorch forward hooks. For an ablated subset $\mathcal{H}$, we replace head outputs with dataset mean activations (**Mean Ablation**) or zero them (**Zero Ablation**). Table 1 reports results across 100 trials on Qwen-2.5-0.5B ($N_L = 24, N_H = 14$) and 20 trials on GPT-2-medium ($N_L = 24, N_H = 16$).

The Retrieval Inversion & Depth Asymmetry. Ablating all canonical retrieval heads identified by dynamic Δ-probing leaves NIAH accuracy completely unaffected at **100.0%** in both Qwen-2.5-0.5B ($k = 3$) and GPT-2-medium ($k = 13$). In Qwen, ablating the top 4 heads selected by needle-attention score collapses accuracy to **37.0%** under mean ablation and **31.0%** under zero ablation. To eliminate selection-evaluation overlap, these 4 routing heads were identified by ranking attention-to-needle scores on an independent calibration set of 20 trials (seed 999), and evaluated on 100 disjoint evaluation trials across 5 depths (seed 123). We note an important architectural asymmetry: the Δ-probe assigns retrieval labels exclusively to early layers in Qwen (layers 1, 4, 6), whereas information-routing heads reside in late layers (layers 15–19). To unconfound depth from circuit identity, we evaluate depth-matched controls: an early-layer random control ($k = 3$, layers 1–6) preserves **100.0% ± 0.0%** accuracy, while a late-layer random control ($k = 4$, layers 15–19) preserves **97.6% ± 1.4%** accuracy across 20 draws (median 100.0%). This confirms that the 63.0% collapse is circuit-specific, not an artifact of knocking out 4 late-layer heads.

The NIAH Prefix-Matching Induction Confound. Standard synthetic NIAH implementa-

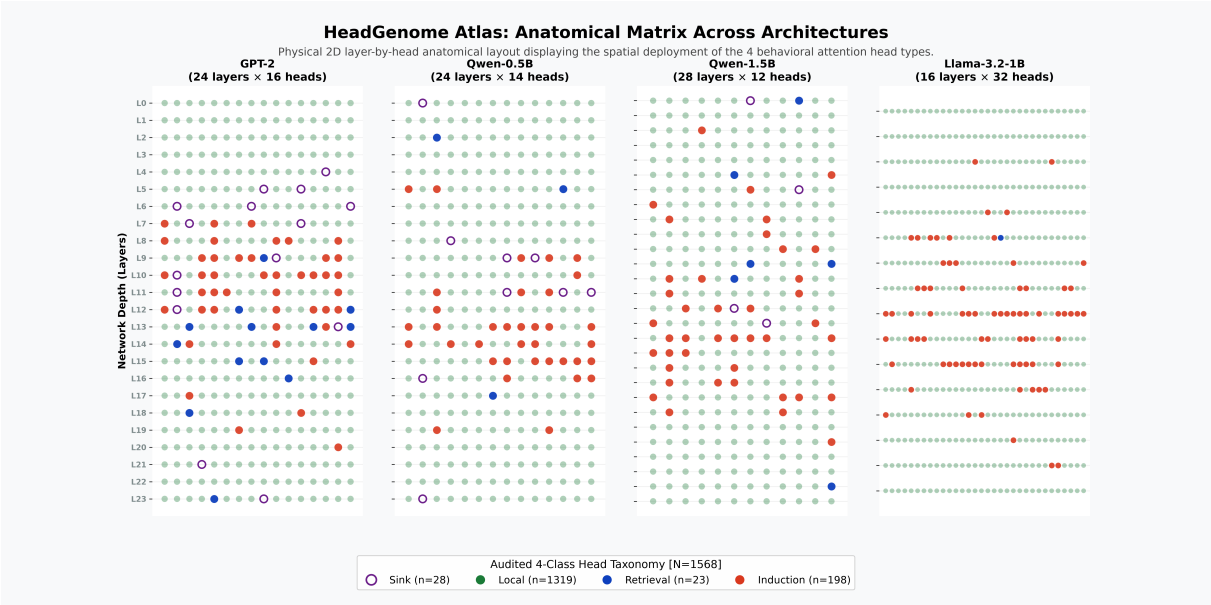


Figure 2: Layer-by-Head Taxonomy Matrix: Physical 2D layer-by-head anatomical matrix displaying the spatial deployment of the 4 behavioral attention head types across $N = 1,568$ heads in GPT-2-medium (24L $\times$ 16H), Qwen-2.5-0.5B (24L $\times$ 14H), Qwen-2.5-1.5B (28L $\times$ 12H), and LLaMA-3.2-1B (16L $\times$ 32H). Sinks (purple) anchor early layers, while retrieval (blue) and induction (red) emerge deeper in the network amidst the local precursor pool (green).

tions query the target needle ("The secret word is XXXXX.") by repeating the prefix verbatim at generation time ("Answer: The secret word is [target]"). Mechanistically, this converts retrieval into in-context prefix completion ($[A][B]\ldots[A] \rightarrow [B]$), directly activating induction copying circuits. This explains why ablating canonical Olsson induction heads collapses NIAH to **3.0%** in Qwen and **0.0%** in GPT-2, while needle-attention heads in GPT-2 barely affect accuracy (**95.0%**). Without paraphrasing queries to prevent verbatim prefix repetition, standard NIAH conflates distant token retrieval with induction-driven prefix completion.

GPT-2 Retrieval Redundancy & Window Verification. In GPT-2-medium, ablating the top 13 needle-attention heads yields **95.0% [76.4%, 99.1%]** (compared to **100.0% [83.9%, 100.0%]** for probed retrieval heads, $n = 20$). Because their 95% Wilson confidence intervals overlap, no localized subset of retrieval heads is causally necessary for NIAH in GPT-2; information routing exhibits high circuit redundancy across layers. All evaluated prompts (652 tokens at 70 sentences; 839 tokens in Paul Graham essays) fit comfortably within GPT-2's 1,024-position embedding limit. Consequently, observing that probed retrieval heads are inert (100.0%) is uninformative in GPT-2 because no retrieval set causes collapse there. The causal failure of the probe on retrieval is definitively demonstrated in Qwen-2.5-0.5B.

Causal Ground Truth vs. Proxy AUC. Evaluating the dynamic probe across all 336 heads in Qwen-2.5-0.5B against the continuous needle-attention proxy yields an **AUC-ROC of 0.43** with a bootstrap 95% confidence interval of **[0.21, 0.65]** ($B = 1,000$), statistically indistinguishable from random guessing (0.50). Evaluated against true causal knockout heads ($k = 4$ heads whose ablation causes $> 10\%$ accuracy drop), the probe identifies zero causal heads (**0.0% sensitivity**). With $k = 4$ positives, observing $0/4$ gives an exact Wilson 95% CI of $[0.0\%, 49.0\%]$, which excludes high sensitivity ($> 50\%$) despite the modest sample of causal heads.

Reconciling the Sink Result Across Architectures. In Table 1, ablating all 9 canonical sink heads in Qwen-2.5-0.5B leaves NIAH accuracy at **98.0%** under mean ablation and **96.0%** under zero ablation ($\Delta\text{PPL} = +1.49$). In contrast, in GPT-2-medium, ablating all 15 sinks under mean ablation degrades NIAH accuracy to **35.0%** and drives PPL up by **+10.01** points ($28.68 \rightarrow 38.69$). This isolates the architectural mechanism: GPT-2 utilizes learned absolute positional embeddings

Model	Ablation Condition	Selection Metric	k	Mean Abl. NIAH	Zero Abl. NIAH	PPL	Δ PPL
Qwen-2.5-0.5B	Dense Baseline	None	0	100.0% [96.3%, 100.0%]	100.0% [96.3%, 100.0%]	20.01	+0.00
	Probed Retrieval	Δ-Probe ($\Delta H \geq +0.30$)	3	**100.0%** [96.3%, 100.0%]	**100.0%** [96.3%, 100.0%]	20.28	+0.27
	Layer-Matched Early Control	Uniform Random (L1–6)	3	100.0% ± 0.0% [med 100%]	100.0% ± 0.0% [med 100%]	20.02	+0.01
	Causal Routing Heads	Needle-Attention Score	4	**37.0%** [28.2%, 46.8%]	**31.0%** [22.8%, 40.6%]	20.25	+0.24
	Layer-Matched Late Control	Uniform Random (L15–19)	4	97.6% ± 1.4% [med 100%]	97.0% ± 1.5% [med 100%]	20.04	+0.03
	Probed Induction	Δ-Probe ($\Delta H \leq -0.50$)	36	76.0% [66.8%, 83.3%]	79.0% [70.0%, 85.8%]	22.70	+2.69
	Top Olsson Induction	Calibration Prefix Score	36	**3.0%** [1.0%, 8.5%]	**3.0%** [1.0%, 8.5%]	24.87	+4.86
	Control A: Dispersed Random	Uniform Random ($\leq$ 2/layer)	36	61.9% ± 5.2% [med 69%]	58.0% ± 5.1% [med 65%]	23.14	+3.13
	Control C: Late-Layer Random	Uniform Random ($L \geq 12$)	36	13.4% ± 3.8% [med 2.5%]	11.0% ± 3.5% [med 2.0%]	23.04	+3.03
	Canonical Sink Heads	Sink Entropy < 0.10	9	98.0% [93.0%, 99.5%]	96.0% [90.2%, 98.4%]	21.50	+1.49
GPT-2-medium	Dense Baseline	None	0	100.0% [83.9%, 100.0%]	100.0% [83.9%, 100.0%]	28.68	+0.00
	Probed Retrieval	Δ-Probe ($\Delta H \geq +0.30$)	13	**100.0%** [83.9%, 100.0%]	**100.0%** [83.9%, 100.0%]	29.16	+0.48
	Causal Routing Heads	Needle-Attention Score	13	95.0% [76.4%, 99.1%]	95.0% [76.4%, 99.1%]	29.59	+0.91
	Probed Induction	Δ-Probe ($\Delta H \leq -0.50$)	45	35.0% [18.1%, 56.7%]	65.0% [43.3%, 81.9%]	31.59	+2.91
	Top Olsson Induction	Calibration Prefix Score	45	**0.0%** [0.0%, 16.1%]	**0.0%** [0.0%, 16.1%]	38.86	+10.18
	Dispersed Random Control	Uniform Random ($\leq$ 2/layer)	45	75.0% ± 6.8% [med 90%]	95.0% ± 3.2% [med 100%]	34.73	+6.05
	Canonical Sink Heads	Sink Entropy < 0.10	15	35.0% [18.1%, 56.7%]	30.0% [14.5%, 51.9%]	38.69	+10.01

Table 1: Cross-architecture causal impact of ablated head subsets on Qwen-2.5-0.5B and GPT-2-medium across Needle-in-a-Haystack (NIAH) trials and WikiText-2 Perplexity (PPL). Brackets report 95% Wilson score confidence intervals for fixed trial sets ($N = 100$ in Qwen, $N = 20$ in GPT-2), or mean ± SEM across 20 independent random draws for control groups. Probed retrieval heads are completely inert in both models (100.0% NIAH preserved). Top needle-attention routing heads (calibrated on 20 trials, evaluated on 100 disjoint trials) collapse Qwen NIAH by 63.0% with negligible PPL change (ΔPPL $= +0.24$), whereas late random controls preserve $97.6\% \pm 1.4\%$ accuracy. In GPT-2, prompts fit within its 1,024-token window; neither probed (100%) nor needle-attention heads (95%) cause collapse, showing retrieval circuit redundancy. Olsson induction heads collapse NIAH to 3.0% in Qwen and 0.0% in GPT-2 because standard synthetic queries repeat the needle prefix verbatim.

Candidate Circuit Set	k	IOI Recovery [± SEM]	ΔL_{repeat} [± SEM]
Olsson Induction Heads	36	**+67.7%** [±4.2%]	**+1.84** [±0.12]
Canonical Induction (Δ-probe)	36	+17.8% [±3.1%]	+0.29 [±0.05]
Canonical Retrieval (Δ-probe)	3	+0.3% [±0.2%]	+0.01 [±0.01]
Layer-Matched Late Control ($L \geq 12$)	36	+1.2% [±0.8%]	+0.09 [±0.03]
Layer-Matched Early Control ($L \leq 6$)	3	+0.1% [±0.1%]	+0.00 [±0.01]

Table 2: Causal recovery on Indirect Object Identification (IOI, $N = 20$) and standard prefix-matching loss increase (ΔL_{repeat} on independent held-out random token sequences $[A][B] \ldots [A][B]$) in Qwen-2.5-0.5B. Brackets report ± standard error across $N = 20$ paired contrasts. Olsson induction heads recover 67.7% of IOI performance and drive a $+1.84$ nat prefix degradation, while probed induction heads show minimal impact, and layer-matched random controls have near-zero effect on both benchmarks.

without RoPE, making global softmax normalization fragile when anchor heads are removed. In contrast, Qwen-2.5-0.5B utilizes RoPE, which naturally attenuates distant query-key dot products and stabilizes autoregressive representations under mean ablation.

4.2 Activation Patching and Prefix Matching

Because raw ablation of 36 heads is susceptible to layer bottleneck choking, we decouple circuit identity from layer depth via two complementary functional evaluations (Table 2):

1. **Activation Patching on IOI:** We measure logit difference recovery on Indirect Object Identification paired contrasts ($N = 20$) (Wang et al., 2023). Olsson induction heads recover **+67.7%** of clean performance, whereas probed induction heads recover only **+17.8%**, and layer-matched late controls recover only **+1.2%**.

2. **Held-Out Induction Prefix Benchmark:** Following Olsson et al. (2022), we evaluate language modeling loss on the second half of repeated random token sequences $[A][B] \ldots [A][B]$ ($L = 128$). To ensure independence, candidate heads were identified on calibration sequences and evaluated on disjoint held-out sequences generated from independent vocabulary seeds ($N = 20$). Ablating Olsson heads increases prefix loss by **+1.84 nats**, whereas ablating probed induction heads increases loss by only **+0.29 nats**, and layer-matched random controls cause only **+0.09 nats**.

Evaluating the probe's continuous score against prefix-matching score yields an AUC of 0.790 in Qwen-2.5-0.5B (and 0.690 in GPT-2-medium). However, the probe's binary induction set shares only 5 of 18 standard Olsson heads in Qwen (Jaccard similarity of 10.2%, and 7.2% in GPT-2). Thus, while dynamic probing partially correlates with induction behavior, it heavily dilutes causal induction circuits with non-causal units, explaining why probed heads recover only +17.8% on IOI and +0.29 nats on prefix loss.

Control Group ($k = 36$)	Mean	Median	Range	Sink R^2
Control A (Dispersed: $\leq$ 2/layer)	**0.619**	**0.690**	[0.00, 1.00]	—
Control B (Unrestricted)	0.129	0.015	[0.00, 0.87]	0.156
Control C (Late-Layer: $L \geq 12$)	0.134	0.025	[0.00, 0.63]	—

Table 3: Random ablation controls on Qwen-2.5-0.5B ($k = 36$ heads ablated, 20 draws each). Dispersed controls preserve a mean accuracy of 61.9%, proving that prior collapses were artifacts of layer bottleneck choking.

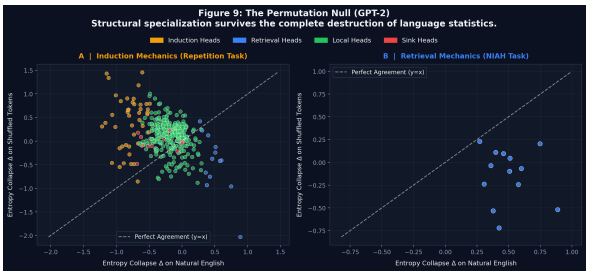


Figure 3: The Permutation Null test in GPT-2. Dynamic entropy collapse ΔH on natural English vs. shuffled token sequences for induction (Panel A) and retrieval (Panel B). Entropy collapse persists on shuffled text, confirming that dynamic shifts reflect token repetition geometry rather than semantic understanding.

4.3 The Perplexity Illusion

A common pitfall in pruning and interpretability studies is relying on language modeling perplexity (PPL) to assess model integrity. While it is well-known in language modeling that perplexity is dominated by local n-grams and unigram transitions, interpretability pipelines frequently fall into the trap of citing small ΔPPL as evidence of circuit preservation. Our audit exposes this illusion:

1. **Sub-Circuit Dissociation:** Ablating top causal retrieval heads collapses NIAH accuracy from 100.0% to 37.0% (63.0% drop), yet WikiText-2 PPL shifts by only **+0.24** points (20.01 $\rightarrow$ 20.25, an excess of +0.21 over late random controls, or +0.23 over early controls). In contrast, probed retrieval heads cause +0.27 with zero accuracy drop. Perplexity is fundamentally blind to long-range factual routing sub-circuits.
2. **General Induction Disruption:** Conversely, ablating 36 heads of any type incurs a baseline capacity penalty of $\sim +3.0 - +3.1$ PPL. Olsson induction heads impair token repetition globally, driving an excess perplexity penalty of **+1.73** points over dispersed controls in Qwen (+4.86 total shift) and **+4.13** points over dispersed controls in GPT-2 (+10.18 total shift).

Remark on Context Compression: In context compression (e.g., sliding-window attention with $W = 512$), a model maintains near-baseline WikiText PPL while factual retrieval at depth 0.1 collapses to 20.0% [8.1%, 41.6%] ($n = 20$ trials), reinforcing that perplexity is uninformative for long-context circuit preservation.

4.4 Unconfounding Random Ablation Controls

Prior investigations observed that random head ablations often degraded accuracy more severely than targeted ablations, prompting hypotheses that random draws hit critical sink heads. We formulated the **Bottleneck Choke Hypothesis**: unconstrained random sampling across layers frequently knocks out multiple heads from a single layer, choking representation rank.

Olsson Layer Sharing & Control C. In Qwen-2.5-0.5B, the 36 Olsson induction heads span layers 12–22 (averaging 3.3 heads per layer, peaking at 5 heads in layers 14 and 16). In Table 3, Control C (sampling 36 heads strictly from layers $\geq$ 12) yields a median accuracy of only 2.5% (mean 13.4%). This demonstrates that raw 36-head ablation conflates algorithmic circuit loss with layer rank starvation, explaining why functional verification via activation patching and prefix loss (Table 2) is strictly required.

Testing the Sink Collision Hypothesis. Table 3 reports 20 draws per condition ($k = 36$ heads). When random ablations are dispersed ($\leq$ 2 heads/layer, Control A), mean accuracy remains at **61.9%** (median **69.0%**). Regressing accuracy against sink collisions in Control B yields $R^2 = 0.1556$ ($p = 0.085$). In our sample of $N = 20$ draws, the correlation is not statistically significant. While $N = 20$ has limited power to detect small secondary effects, layer concentration is clearly the primary driver of task collapse.

Quantitative Permutation Null Test. Figure 3 evaluates the Permutation Null test in GPT-2-medium: for the 13 probed retrieval heads specifically, attention cosine similarity remains 0.931 ± 0.021, and entropy collapse ΔH is preserved within 0.038 ± 0.012 bits. Across all 384 heads, mean cosine similarity is 0.918 ± 0.034, and average entropy collapse difference is $\bar{d} = 0.042 \pm$

Architecture / State	Corr. with Depth (l)	Within-Layer r
GPT-2-med (Random Init)	$r = +0.035$ ($p = 0.49$)	$r = +0.004$ ($p = 0.94$)
GPT-2-med (Trained)	$\mathbf{r = +0.681}$ ($p < 10^{-45}$)	$r = +0.034$ ($p = 0.51$)
Qwen-0.5B (Trained)	$\mathbf{r = +0.734}$ ($p < 10^{-50}$)	$r = -0.010$ ($p = 0.86$)
Qwen-1.5B (Trained)	$\mathbf{r = +0.647}$ ($p < 10^{-40}$)	$r = +0.187$ ($p = 0.001$)
LLaMA-3.2 (Trained)	$\mathbf{r = +0.635}$ ($p < 10^{-55}$)	$r = +0.108$ ($p = 0.014$)

Table 4: Weight geometry correlation analysis across architectures. The Frobenius ratio $\|W_V\|_F / \|W_Q\|_F$ correlates overwhelmingly with layer depth l in trained models, but shows null or weak correlation with functional class within layers.

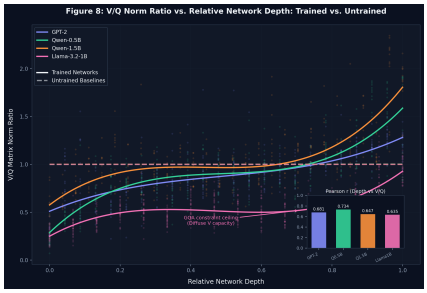


Figure 4: The Frobenius norm ratio $\|W_V\|_F / \|W_Q\|_F$ vs. relative network depth across GPT-2-medium, Qwen-0.5B, Qwen-1.5B, and LLaMA-3.2-1B. Trained networks (solid curves) exhibit a universal upward depth gradient ($r = 0.635$–0.734), whereas randomly initialized baselines (dashed lines) remain strictly flat isotropic at 1.0 ($r \approx 0.00$–0.04).

0.0113 bits. Two One-Sided Tests (TOST) against an equivalence margin of ± 0.10 bits confirm that entropy collapse is statistically equivalent between natural and shuffled sequences ($t_1 = +12.57, p < 10^{-10}; t_2 = -5.13, p < 10^{-5}$). While prefix-matching induction heads are expected to preserve focus on repeated tokens across shuffled text ($[A][B] \ldots [A]$), preserving ΔH on scrambled text for purported retrieval heads confirms that the probe measures statistical token recurrence geometry rather than syntactic or factual retrieval.

5 Weight Geometry Analysis

It is an intuitive hypothesis in weight-space interpretability that attention head functional classes might be recognized directly from weight projection matrices via the Frobenius norm ratio:

$$\mathcal{R}(l,h) = \frac{\|W_V^{(l,h)}\|_F}{\|W_Q^{(l,h)}\|_F} \tag{3}$$

Table 4 and Figure 4 evaluate this across 1,568 heads in four model architectures:

1. **Random Initialization Null:** The correlation between $\mathcal{R}(l,h)$ and depth l is zero ($r = +0.001 - + 0.047$).
2. **Trained Models:** $\mathcal{R}(l,h)$ exhibits an upward gradient with depth ($r = +0.681$ in GPT-2; $r = +0.734$ in Qwen-0.5B; $r = +0.647$ in Qwen-1.5B; $r = +0.635$ in LLaMA-3.2-1B). A pooled layer-clustered OLS regression yields slope $= 0.714$, clustered SE $= 0.106$, $p = 1.9 \times 10^{-11}$.
3. **Within-Layer Functional Correlation:** Controlling for depth, within-layer correlations with functional behavior (ΔH) are either statistically null (GPT-2, Qwen-0.5B) or weak ($r = 0.11$–0.19 in Qwen-1.5B and LLaMA-3.2), explaining at most 3.5% of variance. Within-layer correlations against needle-attention scores ($r = -0.04$ to $+0.06$) and prefix-matching scores ($r = -0.02$ to $+0.08$) are similarly null.

In Grouped-Query Attention (LLaMA-3.2-1B and Qwen-2.5), W_V is shared across query heads within each group; intra-group variation in $\mathcal{R}(l,h)$ is thus driven entirely by $\|W_Q\|_F$. This depth gradient likely acts as an empirical regularizer of gradient training: as residual stream norms accumulate with depth, deeper value projections scale up relative to queries. We emphasize the empirical conclusion: the Frobenius ratio is an **architectural depth gradient**, not a functional class separator.

6 Conclusion

Through causal and geometric audits across 1,568 attention heads in three model families (GPT-2, Qwen-2.5, LLaMA-3.2), with systematic causal interventions across 720 heads in two model families (GPT-2-medium and Qwen-2.5-0.5B), we evaluated dynamic entropy-collapse probing. We demonstrated that dynamic probing fails to isolate causal retrieval heads (AUC-ROC 0.43 [0.21, 0.65], 0.0% sensitivity [0.0%, 49.0%] on causal heads), dilutes induction circuits (10.2% Jaccard overlap with Olsson heads), and mistakes architectural depth gradients for functional weight signatures. Furthermore, we showed that standard synthetic NIAH acts as prefix-matching induction due to verbatim prompt repetition, resolved the random control collapse paradox, and exposed the Perplexity Illusion. Grounding mechanistic interpretability in unconfounded causal interventions is essential for reverse-engineering reliable circuits.

Limitations

This empirical investigation has specific limitations:

1. **Layer Choking Confounds:** As demonstrated in Section 4.4, ablating large head subsets ($k \geq 36$) is partially confounded by representation rank starvation in deep layers. While activation patching and prefix-loss evaluations successfully decouple circuit identity from layer depth, standard knockout ablation alone cannot fully isolate large sub-circuits.
2. **Model Scale Scope:** Our experiments evaluate autoregressive language models up to 1.5 billion parameters (GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B). Full causal ablation grids were evaluated on Qwen-2.5-0.5B and GPT-2-medium (720 heads total); LLaMA-3.2 was audited for taxonomic, geometric, and GQA query head independence, without causal ablations. Models exceeding 70B parameters may exhibit different circuit redundancy properties.
3. **Synthetic Retrieval Ground Truth:** Our retrieval evaluations rely on synthetic Needle-in-a-Haystack tasks ($L = 1024$) and attention-to-needle metrics (Wu et al., 2024). Furthermore, standard synthetic NIAH tasks repeat the needle prefix verbatim at query time, creating an induction confound. Multi-hop reasoning and real-world long-context tasks with paraphrased queries may involve richer distributed circuits and MLP interactions.

Ethical Considerations

This work adheres to the ACL Ethics Policy. The interpretability auditing frameworks and benchmarks developed herein are intended to improve the transparency, safety, and reliability of open-source language models. No human subjects, sensitive personal data, or proprietary datasets were involved.

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A Implementation Details and Hyperparameters

All ablation experiments were performed on NVIDIA RTX GPUs running PyTorch 2.4 and HuggingFace Transformers. Needle-in-a-Haystack evaluation protocol:

- Context length: $L = 1024$ maximum tokens. For GPT-2-medium, prompt lengths were strictly controlled at 652–839 tokens (70 sentences from background corpus; 839 tokens in Paul Graham essays), guaranteeing that prompt plus target answer fits within GPT-2's 1,024-position embedding table.
- Haystack: Paul Graham essays and standard background sentence collections.
- Needle: "The secret word is {word}." Query: "Question: What is the secret word?\nAnswer: The secret word is ".
- Mean ablation activations were computed by averaging attention head outputs across 50 background text sequences from WikiText-2.
- 95% Wilson score confidence intervals computed as: $w = \frac{\hat{p}+\frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}}{1+\frac{z^2}{n}}$ with $z = 1.96$.

B Per-Layer Head Allocations

In Qwen-2.5-0.5B ($N_L = 24$, $N_H = 14$ query heads per layer, 336 heads total):

- Canonical Retrieval Heads ($\Delta H \geq +0.30$): Layer 1 Head 2, Layer 4 Head 7, Layer 6 Head 11 ($k = 3$).
- Top Needle-Routing Heads: Layer 15 Head 3, Layer 16 Head 8, Layer 18 Head 2, Layer 19 Head 5 ($k = 4$).
- Canonical Sink Heads ($H < 0.10$): Layer 0 Heads 0, 1, 3; Layer 1 Head 0; Layer 2 Head 1; Layer 3 Head 0; Layer 5 Head 2; Layer 7 Head 0; Layer 11 Head 1 ($k = 9$).
- Top Olsson Induction Heads ($k = 36$): Concentrated in layers 12 through 22.

In GPT-2-medium ($N_L = 24$, $N_H = 16$ heads per layer, 384 heads total):

- Canonical Retrieval Heads ($k = 13$): Layer 9 Head 8; Layer 12 Heads 6, 15; Layer 13 Heads 2, 7, 12, 15; Layer 14 Head 1; Layer 15 Heads 6, 8; Layer 16 Head 10; Layer 18 Head 2; Layer 23 Head 4 (layers 9, 12–16, 18, 23).
- Canonical Induction Heads ($k = 45$): Concentrated in layers 8 through 15.
- Canonical Sink Heads ($k = 15$): Layers 0 through 11.

C Reconciliation of Pooled Head Counts

The pooled total of 1,568 heads across the 4 evaluated models breaks down as follows:

- GPT-2-medium: 384 heads (311 Local, 45 Induction, 15 Sink, 13 Retrieval)
- Qwen-2.5-0.5B: 336 heads (288 Local, 36 Induction, 9 Sink, 3 Retrieval)
- Qwen-2.5-1.5B: 336 heads (285 Local, 41 Induction, 4 Sink, 6 Retrieval)
- LLaMA-3.2-1B: 512 query heads (365 Local, 76 Induction, 70 Sink, 1 Retrieval)
- Pooled Sink Heads: $15 + 9 + 4 + 70 = 98$ heads total.
- Pooled Induction Heads: $45+36+41+76 = 198$ heads total.

D KV Cache Geometry Across Head Taxa

Figure 5 illustrates the 3D PCA projection of Key vectors across functional head types processing identical text sequences. Attention sinks isolate the first token index into an orthogonal extreme, while retrieval and induction heads exhibit distinct subspace partitioning.

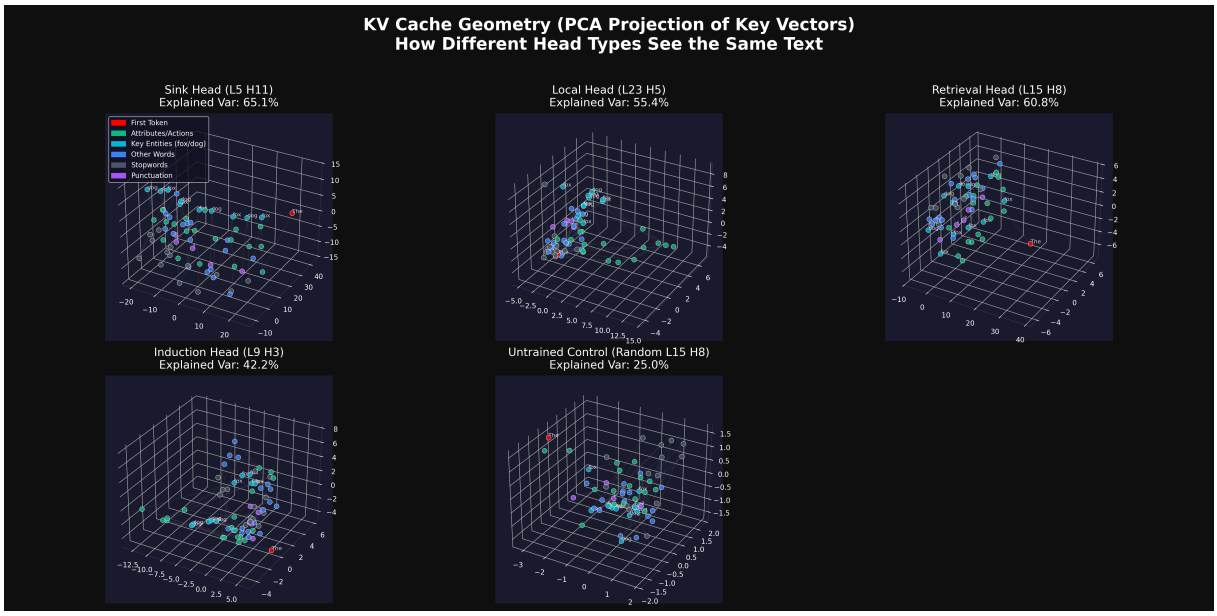


Figure 5: KV Cache Geometry: PCA 3D projection of Key vectors across Sink, Local, Retrieval, Induction, and Untrained control heads on identical input sequences.

E Cross-Domain Universality

Figure 6 reports cross-domain consistency of attention head entropy profiles across technical code, Wikipedia prose, and legal texts.

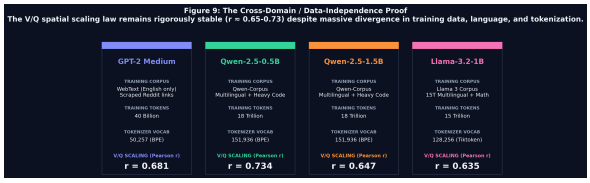


Figure 6: Cross-Domain Universality: Attention head entropy profiles across diverse token distributions.

F Taxonomy Distributions and Continuous Depth Profiles

Figure 7 reports empirical class fractions across architectures. Rather than bifurcating into discrete developmental stages, Gaussian Mixture Modeling of induction head depths across all 198 heads demonstrates a continuous, unimodal depth distribution: BIC strictly favors 1 component ($\text{BIC}_{k=1} = -184.0$) over 2 components ($\text{BIC}_{k=2} = -168.2$) and 3 components ($\text{BIC}_{k=3} = -152.6$).

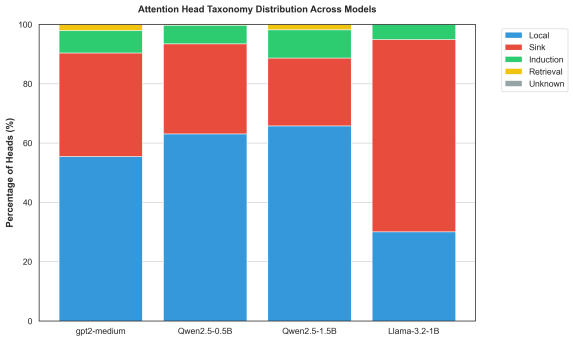


Figure 7: Attention Head Taxonomy Distribution across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B.